

Module 03: Two-Phase Locking Variants (Strict, Rigorous, Conservative)

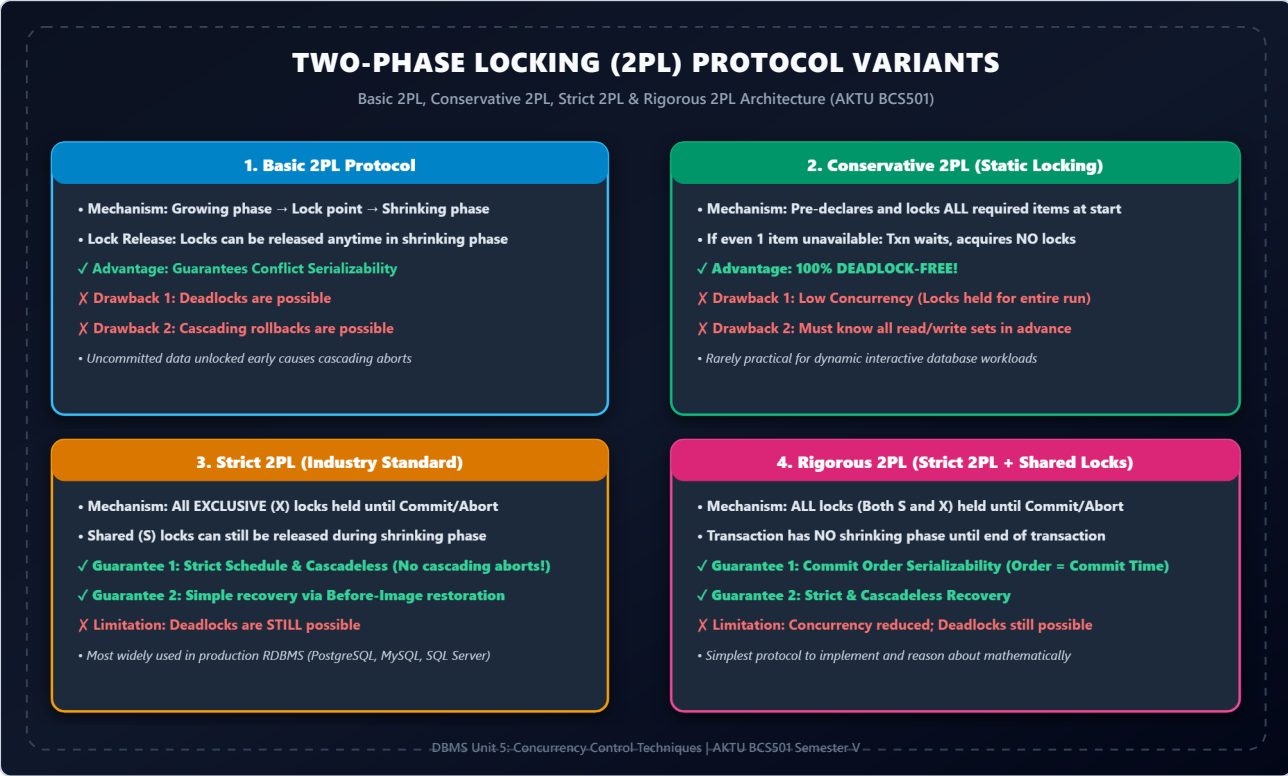


Figure 3.1: Basic, Conservative, Strict and Rigorous 2PL Protocol Architecture

1. Why 2PL Variants are Necessary

Basic 2PL guarantees Conflict Serializability, but allows Deadlocks and Cascading Aborts. Variants solve these operational bottlenecks.

2. Comparative Breakdown of Variants

Variant	Lock Release Timing	Deadlock-Free?	Cascadeless?
Basic 2PL	Shrinking phase	No	No
Conservative 2PL	Pre-locks all items at start	YES	No
Strict 2PL	Exclusive locks held till Commit/Abort	No	YES (Strict)
Rigorous 2PL	All locks (S & X) held till Commit/Abort	No	YES (Strict)

Industry Standard: Strict 2PL

Modern RDBMS (PostgreSQL, MySQL InnoDB, Oracle) employ Strict 2PL because it guarantees strict recoverability and simple before-image rollback while maintaining high concurrency.